BBA SEMESTER III
PYTHON FOR DATA ANALYTICS FOR BUSINESS
CASE STUDY: EVALUATING PROMOTIONAL CAMPAIGN PERFORMANCE

Case Background

BrightWave Retail runs promotional activities to attract customers and increase sales. Its campaigns include discount offers, festival campaigns, loyalty programmes, new product launches, and referral offers. Promotions are delivered through Instagram, Facebook, email, SMS, and in-store activities across four regions.

The marketing manager wants to know which channels and campaign types perform well, how much the business spends, how many people respond to promotions, and how campaign performance changes over time. The campaign records may contain missing values, duplicate rows, and inconsistent entries.

You are working as a junior marketing data analyst. Use Python to clean, analyse, and visualise the campaign data, then present practical recommendations to help the manager plan future promotional activities.

Instructions to Students

- Complete all 10 questions in a Jupyter Notebook.
- Use Pandas, NumPy, Matplotlib, and Seaborn wherever appropriate.
- Include Python code, outputs, charts, and brief interpretations.
- Write each question as a separate heading in the notebook.
- Explain findings in simple business language and support recommendations with results, within the notebook as Text.
- Submit the .ipynb file with your name, roll number, and section. Ensure the notebook runs from beginning to end.

Dataset

Use the accompanying Excel workbook, worksheet 'Promotion Data'. It contains 150 campaign records and deliberately includes some missing values, inconsistent text entries, and duplicate records for cleaning practice.

Variable	Description	Variable
Campaign_ID	Unique campaign record identifier	Campaign_Date: date of campaign
Campaign_Type	Type of promotion	Channel: Instagram, Facebook, Email, SMS, In-store
Region	North, South, East, West	Product_Category: promoted product category
Spend	Campaign expenditure	Impressions: times promotion was seen
Clicks	Clicks or responses	Leads: potential customers showing interest
Conversions	Purchases or completed target actions	Revenue: revenue attributed to campaign

Questions

Q1. Loading and Exploring the Dataset

1. Load the Excel dataset into a Pandas DataFrame.
2. Display the first five and last five records.

3. Find the number of rows and columns, display column names and data types, and identify numerical and categorical columns.

Q2. Data Cleaning and Preprocessing

4. Identify missing values in each column.
5. Identify and remove duplicate records where appropriate.
6. Correct inconsistent entries in Channel, Campaign_Type, Region, and Product_Category.
7. Convert Campaign_Date to the correct date format.
8. Briefly explain why cleaning is important before analysis.

Q3. Creating and Transforming Variables

9. Create a Month column from Campaign_Date.
10. Create a Click_Through_Rate (CTR) column: Clicks ÷ Impressions × 100.
11. Create a Conversion_Rate column: Conversions ÷ Clicks × 100, wherever appropriate.
12. Create a Return_on_Ad_Spend (ROAS) column: Revenue ÷ Spend, wherever appropriate. Display the updated DataFrame and explain the usefulness of these measures.

Q4. Descriptive Analytics Using NumPy and Pandas

13. Calculate total and average Spend, total Impressions, Clicks, Leads, Conversions, and Revenue.
14. Calculate mean, median, minimum, maximum, and standard deviation for suitable numerical columns.
15. Use NumPy to calculate descriptive statistics for Revenue or Spend.
16. Interpret what the results suggest about campaign performance.

Q5. Campaign-Type Analysis

17. Calculate total Spend, total Conversions, and total Revenue for each Campaign_Type.
18. Identify the campaign type with the highest Revenue and the highest number of Conversions.
19. Present a summary table and write two business observations.

Q6. Visualising Campaign Performance Using Matplotlib

20. Create a bar chart showing total Revenue by Channel.
21. Create a line chart showing monthly Revenue trends.
22. Create a histogram showing the distribution of campaign Spend.
23. Add suitable titles, axis labels, and readable tick labels. Interpret each graph in two or three sentences.

Q7. Visualising Campaign Metrics Using Seaborn

24. Create a box plot showing Revenue across different Channels.
25. Create a heatmap showing correlations among suitable numerical variables, such as Spend, Impressions, Clicks, Conversions, and Revenue.
26. Explain what the plots reveal about variation, potential outliers, and relationships among campaign metrics.

Q8. Analysing Promotion Channels Through Visualisation

27. Use Pandas to count campaigns by Channel.
28. Create a Seaborn count plot or Matplotlib bar chart comparing the number of campaigns across channels.
29. Where appropriate, display the campaign counts on the chart.
30. Identify the most frequently used channel and suggest one action the marketing manager could consider.

Q9. Comparing Campaign Results Across Channels and Months

31. Create a grouped bar chart showing monthly Revenue for each Channel.
32. Create a stacked bar chart comparing each Channel's contribution to monthly Revenue.
33. Use Pandas grouping and aggregation to prepare the data for both charts.
34. Add suitable titles, axis labels, legends, and readable labels.
35. Identify the month with the highest overall Revenue and the channel with the highest Revenue across the period.

36. Write three business observations based on the visualisations.

Q10. Business Insights and Recommendations

37. Present three important findings from your descriptive analytics and visualisations.
38. Identify one problem or opportunity in the promotional activities.
39. Recommend three practical actions to improve future promotional campaigns.
40. Explain how visualisation can help the marketing manager monitor performance and make decisions. Support recommendations with your results.

Outcomes

- Python code and outputs for all 10 questions.
- Data cleaning and preprocessing steps.
- Descriptive statistics and derived campaign performance measures.
- Relevant Matplotlib and Seaborn visualisations.
- Brief interpretations and business recommendations.

Evaluation will consider code correctness, appropriate use of analytical tools, data-cleaning steps, visualisation quality, interpretation, and the relevance of business recommendations.